

1.1 Areas Between Curves Worksheet

1. Find the area of the region enclosed by the curves $y = x^2$ and $y = 2x - x^2$.

Find Intersections:

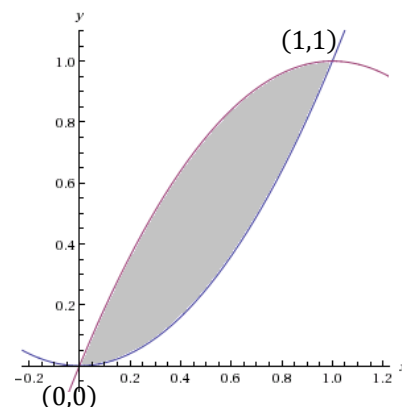
$$x^2 = 2x - x^2 \rightarrow 2x^2 - 2x = 0 \rightarrow 2x(x-1) = 0 \rightarrow x = 1 \text{ or } x = 0$$

Setup:

$$A = \int_0^1 [(2x - x^2) - x^2] dx$$

Evaluate:

$$\begin{aligned} A &= \int_0^1 [(2x - x^2) - x^2] dx = \int_0^1 [2x - 2x^2] dx = \left[x^2 - \frac{2}{3}x^3 \right]_0^1 \\ &= \left[1 - \frac{2}{3} \right] - [0] = \frac{1}{3} \end{aligned}$$



2. Let R be the finite region bounded by the curves $y = 5 \tan(x)$, $y = 5$ and $x = 0$. Set up, but do not evaluate definite integrals which represent the area of the region R . Use your calculator to verify that both integrals produce the same value.

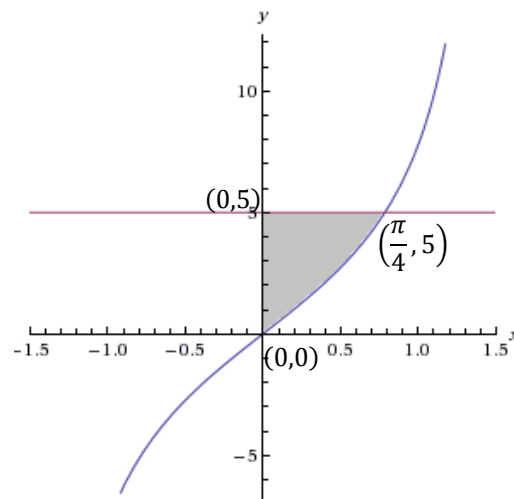
Find Intersections:

$$5 \tan(x) = 5 \rightarrow \tan(x) = 1 \rightarrow x = \frac{\pi}{4}$$

- (a) Integrate with respect to x .

Setup:

$$A = \int_0^{\pi/4} [5 - 5 \tan(x)] dx$$



- (b) Integrate with respect to y .

Solve equations for x :

$$x = \tan^{-1}\left(\frac{y}{5}\right)$$

Setup:

$$A = \int_0^5 \left[\tan^{-1}\left(\frac{y}{5}\right) - 0 \right] dy$$

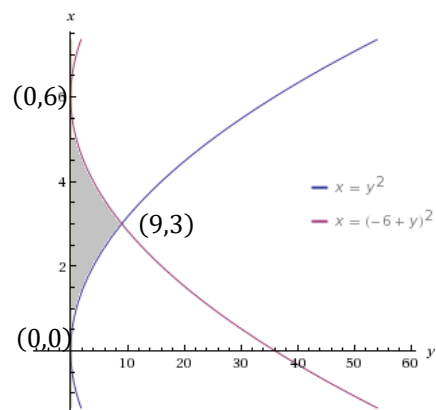
3. Let R be the finite region bounded by the curves $x = y^2$, $x = (y-6)^2$, and $x = 0$. Set up, but do not evaluate definite integrals which represent the area of the region R . Use your calculator to verify that both integrals produce the same value.

Find Intersections:

$$y^2 = (y-6)^2 \rightarrow y^2 = y^2 - 12y + 36 \rightarrow y = 3$$

$$0 = y^2 \rightarrow y = 0$$

$$0 = (y-6)^2 \rightarrow y = 6$$



- (a) Integrate with respect to x .

Solve equations for y :

$$x = y^2 \rightarrow y = \pm\sqrt{x} \rightarrow y = \sqrt{x}$$

$$x = (y-6)^2 \rightarrow y = \pm\sqrt{x} + 6 \rightarrow y = -\sqrt{x} + 6$$

Setup:

$$A = \int_0^9 [-\sqrt{x} + 6 - \sqrt{x}] dx$$

- (b) Integrate with respect to y .

Setup:

Two integrals are needed here:

$$A = \int_0^3 [y^2 - 0] dy + \int_3^6 [(y-6)^2 - 0] dy$$

4. Let R be the finite region bounded by the curves $y = 2\ln(x)$, $y = 3\ln(x)$, and $y = 6$.

- (a) Determine the area of R by evaluating a definite integral with respect to y .

Find Intersections:

$$2\ln(x) = 3\ln(x) \rightarrow \ln(x) = 0 \rightarrow x = 1$$

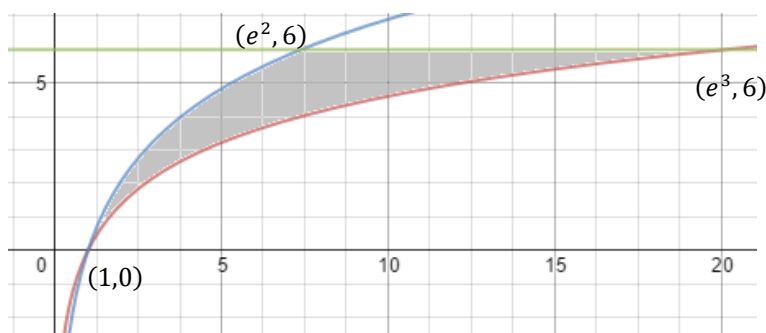
$$2\ln(x) = 6 \rightarrow x = e^3$$

$$3\ln(x) = 6 \rightarrow x = e^2$$

Solve equations for x :

$$y = 2\ln(x) \rightarrow x = e^{y/2} \quad (\text{bottom})$$

$$y = 3\ln(x) \rightarrow x = e^{y/3} \quad (\text{top})$$



Setup:

$$A = \int_0^6 (e^{y/2} - e^{y/3}) dy$$

Evaluate:

$$A = \int_0^6 (e^{y/2} - e^{y/3}) dy = (2e^{y/2} - 3e^{y/3}) \Big|_0^6 = (2e^3 - 3e^2) - (2e^0 - 3e^0) = 2e^3 - 3e^2 + 1$$

(b) Set up, but do not evaluate, definite integral to represent the area of R by evaluating by integrating with respect to x . Use your calculator to see that this integral is equivalent to the one above.

Two integrals are needed here:

Setup:

$$A = \int_1^{e^2} [3 \ln(x) - 2 \ln(x)] dx + \int_{e^2}^{e^3} [6 - 2 \ln(x)] dx$$

5. Let R be the finite region bounded by the curves $f(x) = -2x + 14$ & $g(x) = \frac{20}{x}$.

(a) Determine the area of R by integrating with respect to x .

Find Intersections:

$$-2x + 14 = \frac{20}{x}$$

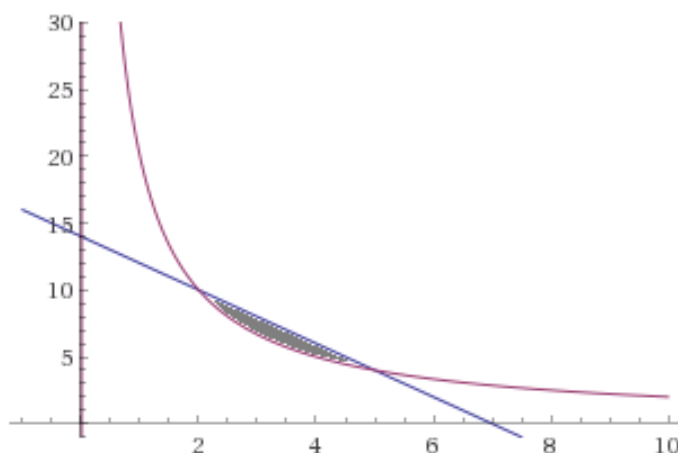
$$\Leftrightarrow -2x^2 + 14x = 20$$

$$\Leftrightarrow x^2 - 7x + 10 = 0$$

$$\Leftrightarrow (x-2)(x-5) = 0$$

$$\Leftrightarrow x = 2 \text{ or } x = 5$$

Points of Intersection: $(2, 10)$ & $(5, 4)$



Setup:

$$A = \int_2^5 \left(-2x + 14 - \frac{20}{x} \right) dx$$

Evaluate:

$$\begin{aligned} A &= \int_2^5 \left(-2x + 14 - \frac{20}{x} \right) dx = \left(-x^2 + 14x - 20 \ln|x| \right) \Big|_2^5 = (-25 + 70 - 20 \ln(5)) - (-4 + 28 - 20 \ln(2)) \\ &= 21 - 20 [\ln(5) - \ln(2)] = 21 - 20 \ln(2.5) \end{aligned}$$

(b) Determine the area of R by integrating with respect to y .

Solve equations for x :

$$y = -2x + 14 \rightarrow x = 7 - \frac{y}{2}$$

$$y = \frac{20}{x} \rightarrow x = \frac{20}{y}$$

Setup:

$$A = \int_4^{10} \left(7 - \frac{y}{2} - \frac{20}{y} \right) dy$$

Evaluate:

$$\begin{aligned} A &= \int_4^{10} \left(7 - \frac{y}{2} - \frac{20}{y} \right) dy = \left(7y - \frac{y^2}{4} - 20 \ln|y| \right) \Big|_4^{10} = (70 - 25 - 20 \ln(10)) - (28 - 4 - 20 \ln(4)) \\ &= 21 - 20[\ln(10) - \ln(4)] = 21 - 20 \ln(2.5) \end{aligned}$$

6. Find the of the region bounded by the curves $y = \frac{x}{\sqrt{x^2 + 1}}$ and $y = x^4 - x$.

(Use your calculator to help find points of intersection and round decimals to 2 places)

Find Intersections:

$$\frac{x}{\sqrt{x^2 + 1}} = x^4 - x \rightarrow x \approx 1.18$$

Setup:

$$A \approx \int_0^{1.18} \left[\frac{x}{\sqrt{x^2 + 1}} - (x^4 - x) \right] dx$$

Evaluate:

$$A \approx \int_0^{1.18} \left[\frac{x}{\sqrt{x^2 + 1}} - (x^4 - x) \right] dx = \int_0^{1.18} \frac{x}{\sqrt{x^2 + 1}} dx - \int_0^{1.18} (x^4 - x) dx$$

Let $u = x^2 + 1$ and thus $du = 2x dx \rightarrow \frac{1}{2} du = x dx$.

Our limits of integration become: $u(0) = (0)^2 + 1 = 1$ & $u(1.18) = (1.18)^2 + 1 \approx 2.39$.

$$A \approx \frac{1}{2} \int_1^{2.9} \frac{1}{\sqrt{u}} du - \int_0^{1.18} (x^4 - x) dx = u^{1/2} \Big|_1^{2.39} - \left(\frac{x^5}{5} - \frac{x^2}{2} \right) \Big|_0^{1.18} = [\sqrt{2.39} - 1] - \left[\frac{(1.18)^5}{5} - \frac{(1.18)^2}{2} \right] \approx 0.78$$

